Experiments: Part II

PS200C Quant Methods in Social Science, Spring 2025

Chad Hazlett

UCLA

chazlett@ucla.edu

Some Important Topics in Experimental Design and Analysis

A brief introduction to some common topics:

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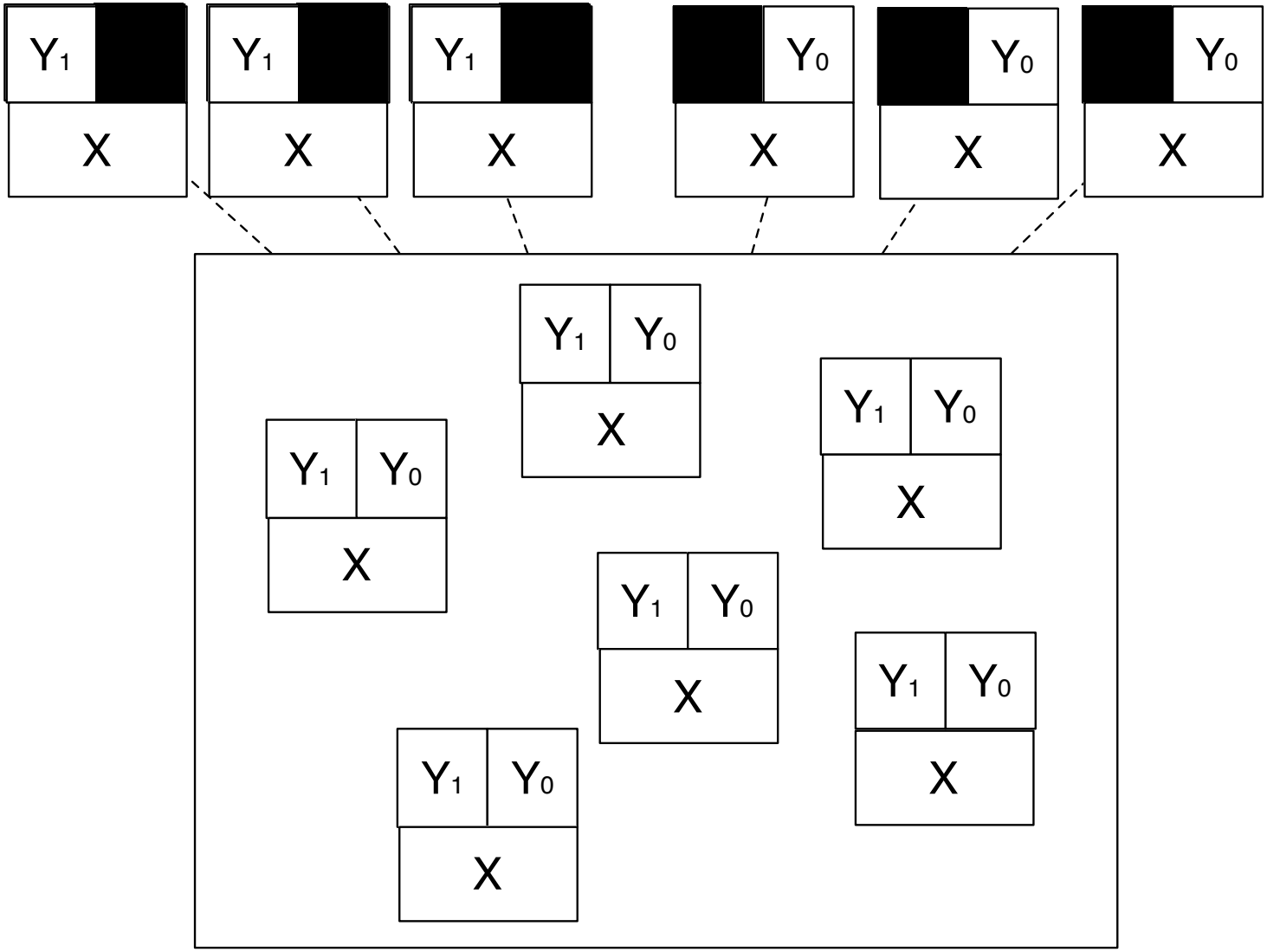
- Covariate adjustment
- Blocking
- Emphasizing design stage

Covariates and Experiments



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- If this is not the case, then one of two possibilities:
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  - finite sample bad luck
- Good idea to (i) check "balance", and/or (ii) adjust for chance imbalances to at least improve efficiency.

Regression with Covariates

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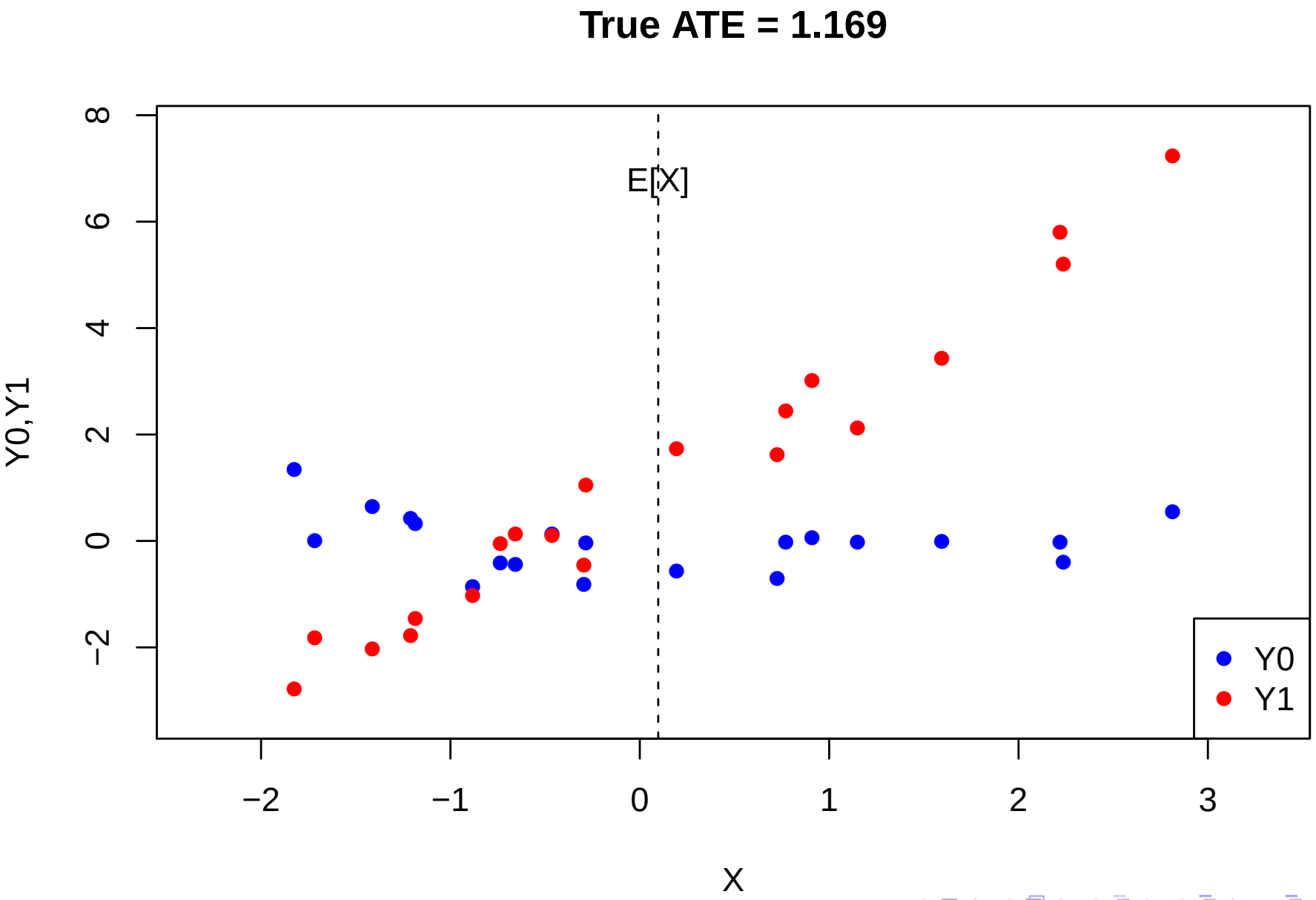
- But, do not control for **post-treatment** covariates!
- This is an over-simplification: sometimes post-tx variables are okay, sometimes your QOI involves post-tx control, and sometimes pre-tx variables are problematic...we will need DAGs for that.

Regression Adjustment for Experiments



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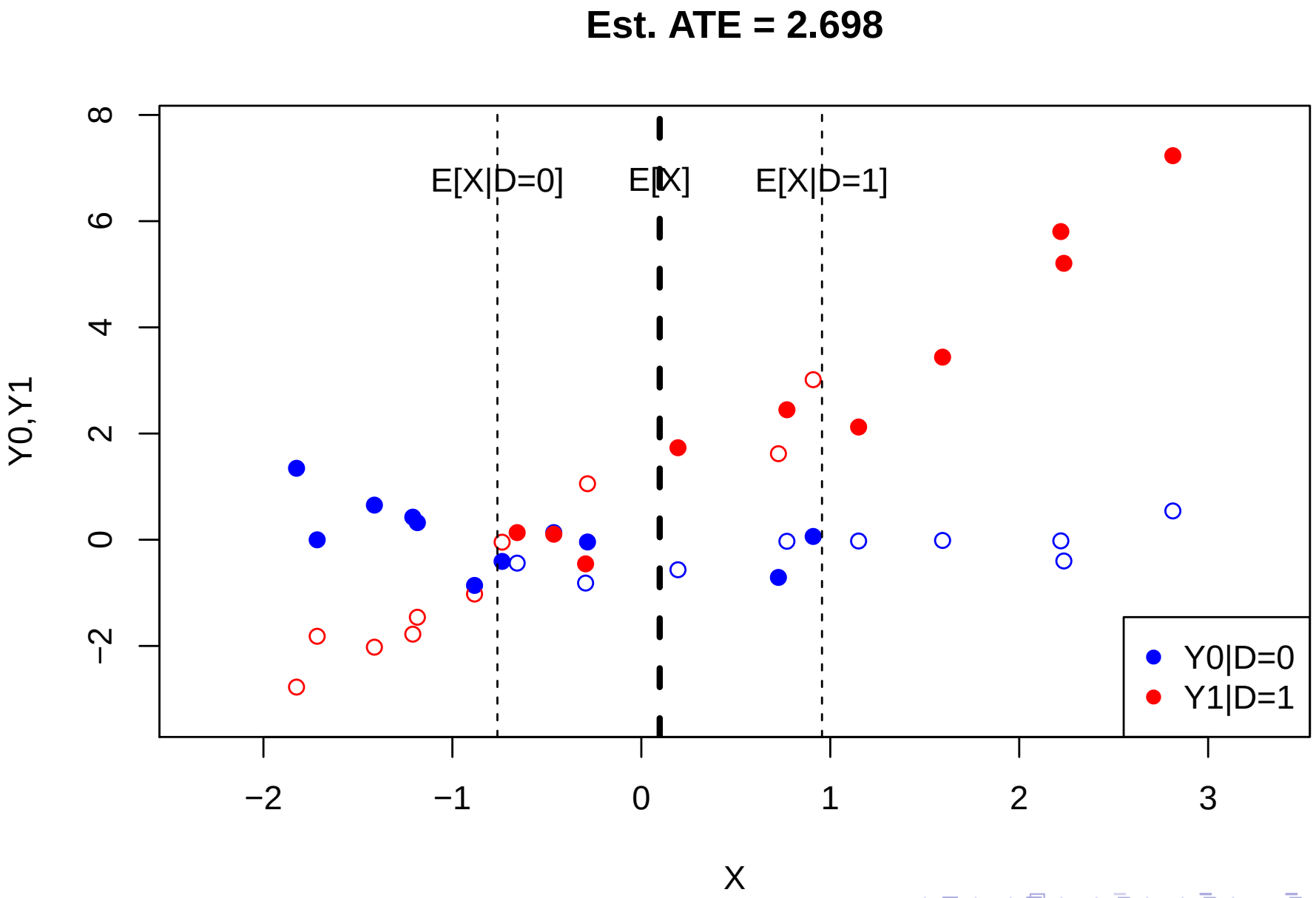
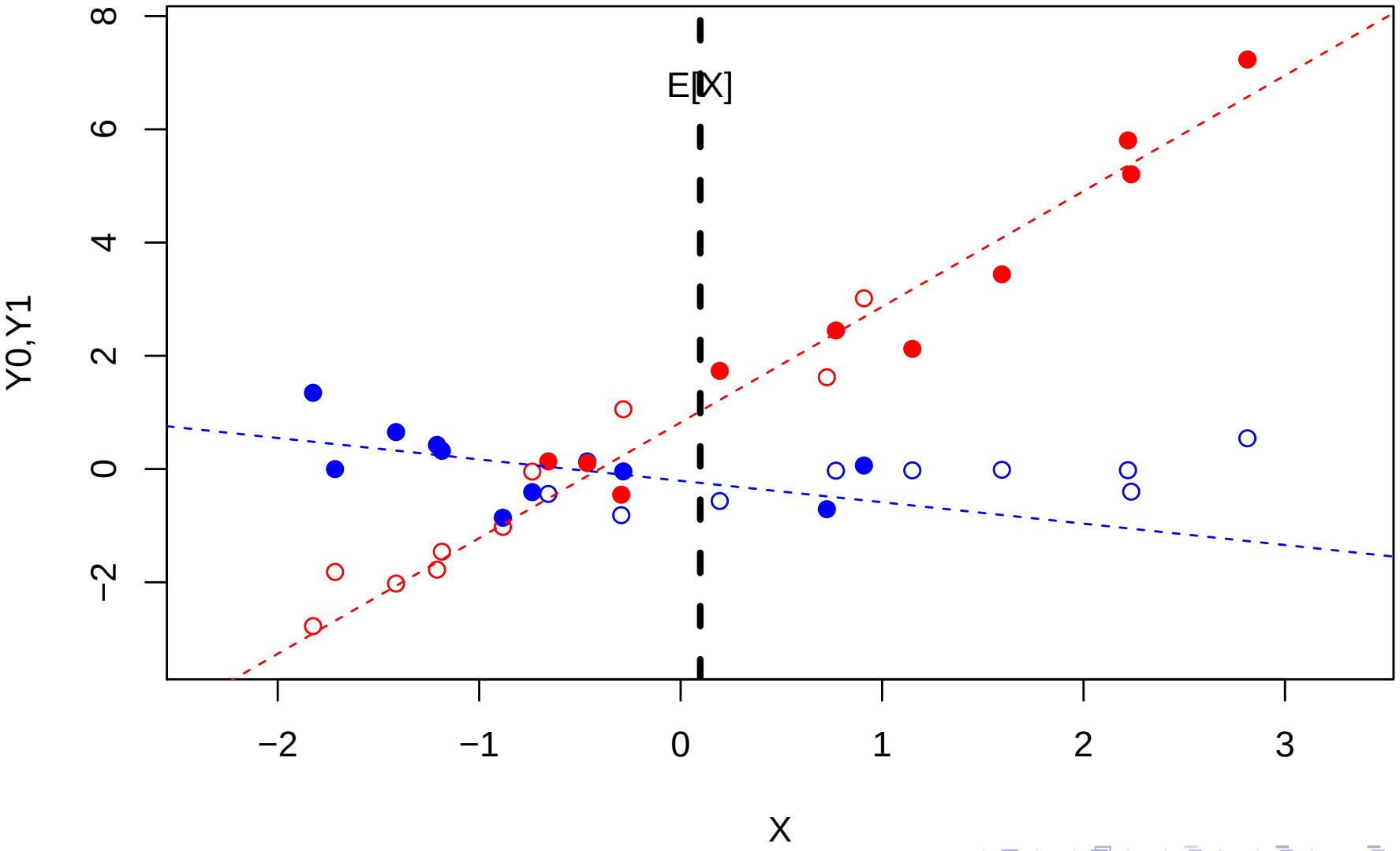


Regression Adjustment For Experiments

Consider regression lines (interacting with D)



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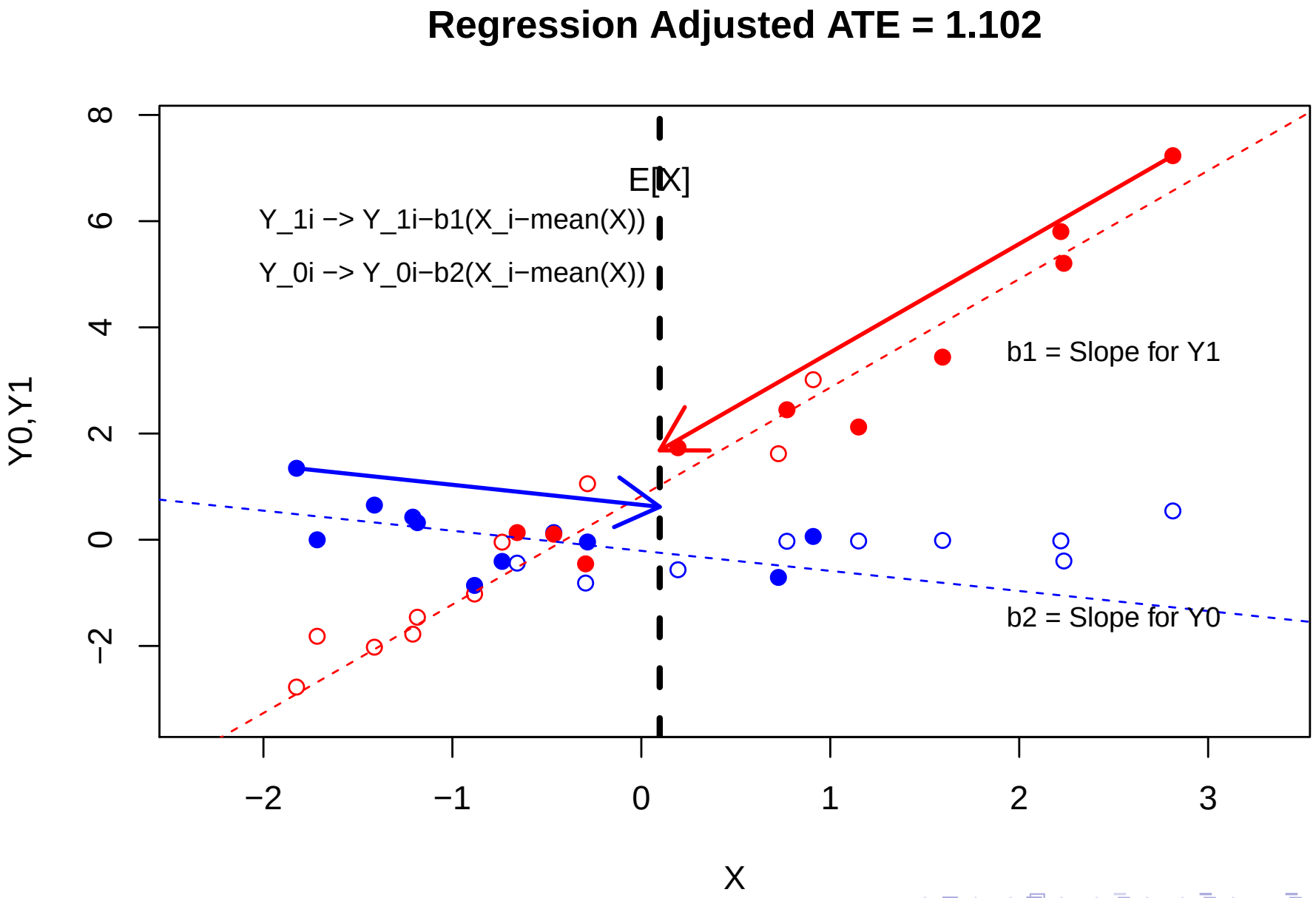

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Better still, (see this working paper):

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- this is also equivalent to some other methods you may hear about (g-computation, t-learner, regression imputation)

Why the weird demeaning business in this model?

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- model could be used to give you $ATE(X)$ but conveniently, $\mathbb{E}_x[ATE(X)] = \tau_{Lin}$.
- allowing this interaction is a good idea, emerging as standard practice

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All together, this gives the Frisch-Waugh-Lovell (FWL) theorem:

$$\beta_k = \frac{Cov(\tilde{Y}_i, \tilde{X}_{ki})}{Var(\tilde{X}_{ki})}$$

Where,

- $\tilde{X}_{ki}$ is the residual from a regression of X_{ki} on all other covariates
- $\tilde{Y}_i$ may have all other X partialled out as well or you can use the original Y_i.

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If our "X_k" of interest is the D:

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- What does this suggest about the effect of adding covariates to the simple regression of Y on D?

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- Helpful both to protect against chance imbalance and improve efficiency.
- Since covariates are balanced by design, results typically not model dependent
- Best reason to be skeptical of adjustment: fishing.
  - always report unadjusted result first
  - best if adjustment strategy pre-specified

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  - Simple random assignment into two treated/control will place $\{2, 2\}$ and $\{8, 8\}$ together in the same group one-third of the time

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or more generally, if there are J strata or blocks, then

$$\hat{\tau}_B = \sum_{j=1}^{J} \frac{N_j}{N} \hat{\tau}_j$$

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If (empirical) $Pr(D = 1)$ changes by block, OLS differently weights the blocks.

We can correct for this by going back to the Lin-style (interacted) regression!'

- Again see this working paper
- Turns out to be a better approach here than demeaning, but that is in the weeds.
- Another proposal is to weight each observation so that block-treatment probabilities become the same. Let

$$p_{ij} \equiv Pr(D_i = 1 | \text{ block for unit i})$$

$$w_{ij} = \left(\frac{1}{p_{ij}}\right) D_i + \left(\frac{1}{1 - p_{ij}}\right)(1 - D_i)$$

...which turns out to be exactly the same thing as the interactive regression

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where R_D^2 is R^2 from regression of D on all block indicators and a constant.

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- If blocks explain outcomes strongly, how will $\widehat{\sigma}^2_{\varepsilon*}$ compare to $\widehat{\sigma}^2_\varepsilon$?
- Adding DOF adjustment, we end up asking whether $\frac{SSR_{\widehat{\varepsilon*}}}{n-k-1} < \frac{SSR_{\widehat{\varepsilon}}}{n-2}$

Fair Trade Labeling Experiment (Hainmueller et al, 2012)

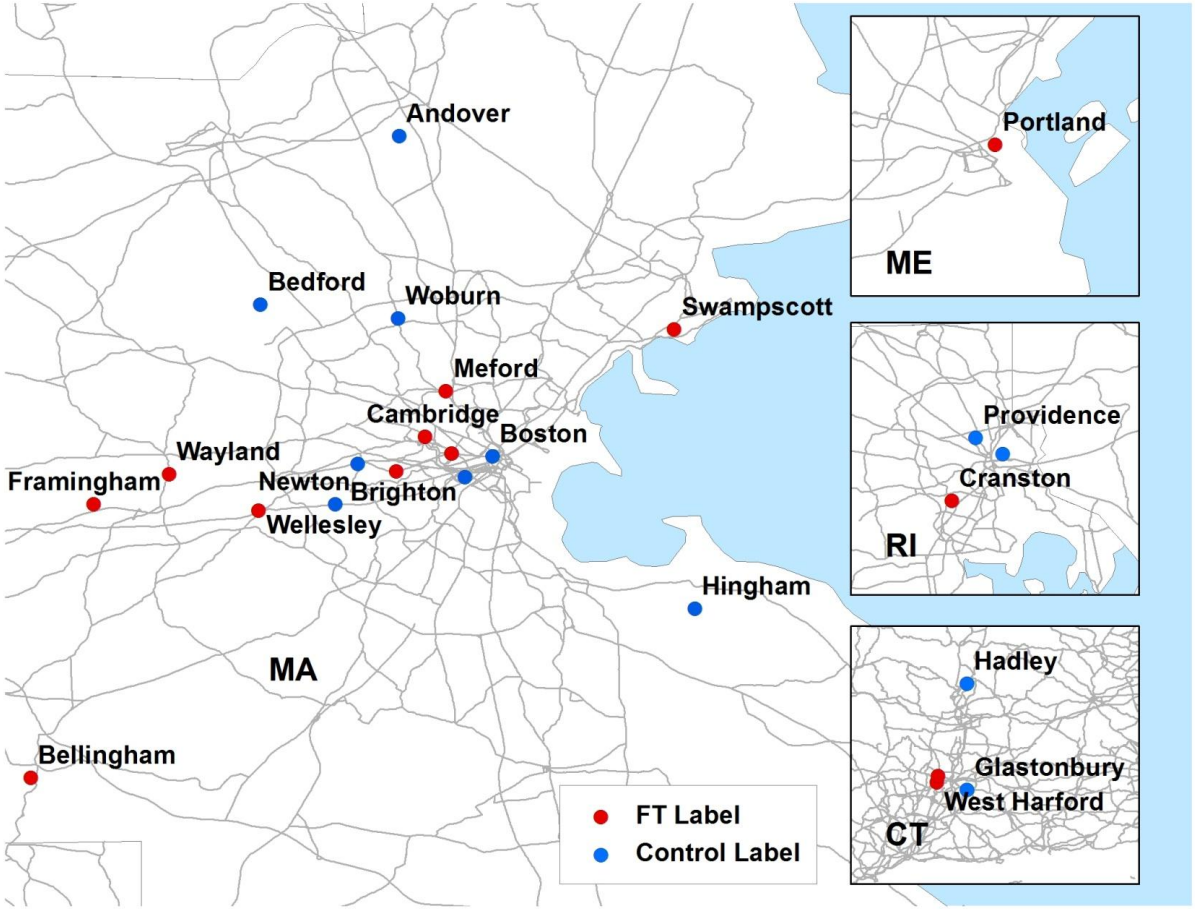
Label Experiment

Treatment

Control

Example: Fair Trade Labeling Experiment

Matched Pairs: Phase 1



Example: Fair Trade Labeling Experiment

```
> d <- read.dta("FTdata.dta")
> head(d)
  store pair FTweek lnsalesd
1     1    1      1     3.20
2     4    1      0     2.77
3     6    2      1     4.18
4     9    2      0     4.04
5    21    3      1     4.30
6    24    3      0     3.93
```

Example: Fair Trade Labeling Experiment

```
> cr.out <- lm(lnsalesd~FTweek,data=d)
> coeftest(cr.out,vcov = vcovHC(cr.out, type = "HC1"))

t test of coefficients:

            Estimate Std. Error t value Pr(>|t|)
(Intercept)  4.35000    0.16079 27.0537   <2e-16 ***
FTweek       0.12385    0.21424  0.5781   0.5686
--
```

Example: Fair Trade Labeling Experiment

```
> br.out <- lm(lnsalesd~FTweek+as.factor(pair),data=d)
> coeftest(br.out,vcov = vcovHC(br.out, type = "HC1"))

t test of coefficients:

                   Estimate Std. Error t value  Pr(>|t|)
(Intercept)        2.923077   0.162144 18.0277 4.671e-10 ***
FTweek             0.123846   0.060176  2.0581 0.0619840
as.factor(pair)2   1.125000   0.159549  7.0511 1.335e-05 ***
as.factor(pair)3   1.130000   0.204440  5.5273 0.0001304 ***
as.factor(pair)4   1.145000   0.231925  4.9369 0.0003439 ***
as.factor(pair)5   1.280000   0.161773  7.9123 4.208e-06 ***
as.factor(pair)6   1.410000   0.169987  8.2948 2.591e-06 ***
as.factor(pair)7   1.575000   0.203689  7.7324 5.317e-06 ***
as.factor(pair)8   1.585000   0.277319  5.7154 9.675e-05 ***
as.factor(pair)9   1.610000   0.169987  9.4713 6.420e-07 ***
as.factor(pair)10  1.795000   0.165195 10.8660 1.450e-07 ***
as.factor(pair)11  1.810000   0.169987 10.6479 1.810e-07 ***
as.factor(pair)12  2.015000   0.164183 12.2729 3.763e-08 ***
as.factor(pair)13  2.070000   0.160298 12.9134 2.127e-08 ***
--
```

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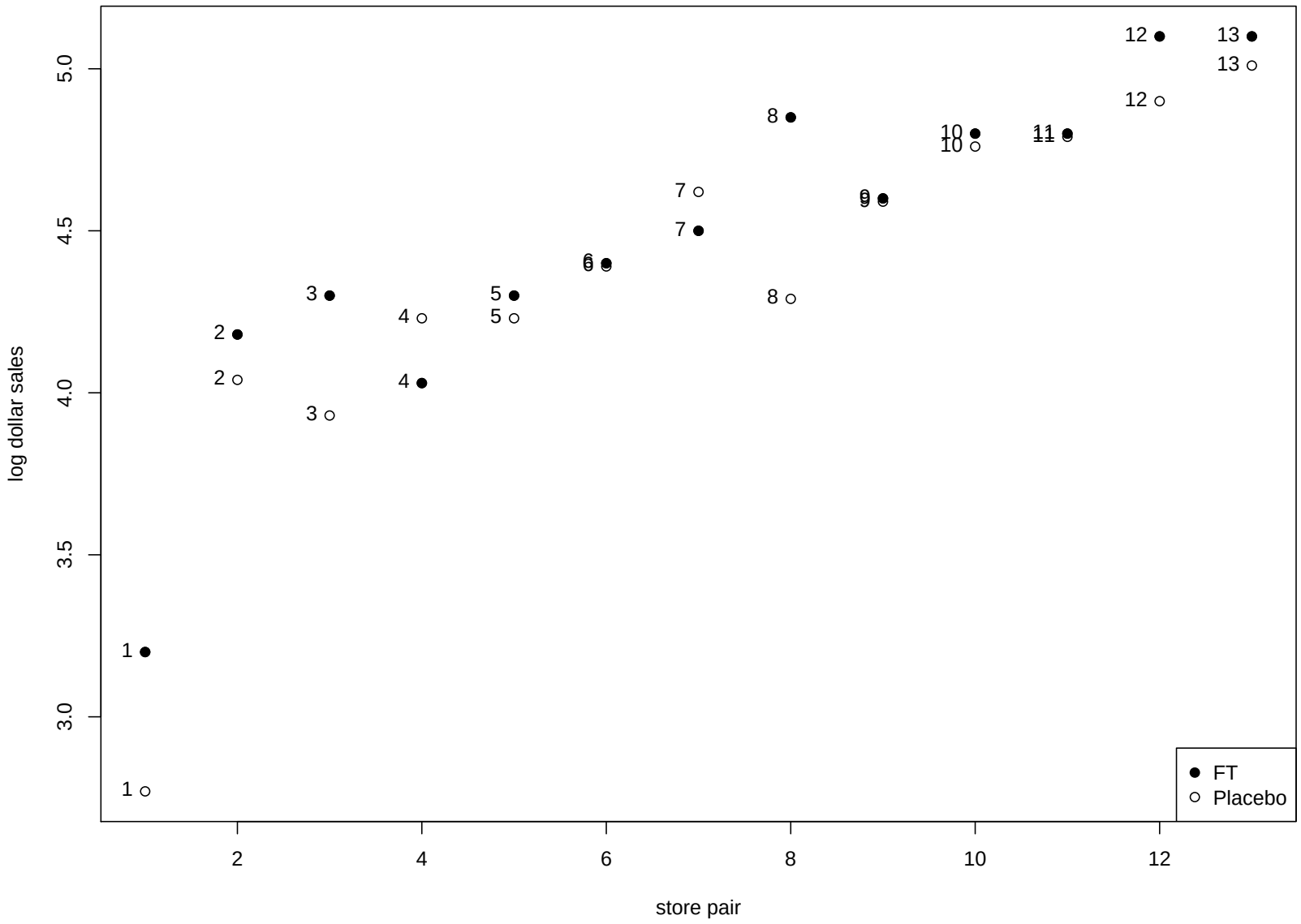

How much is explained by blocks?

```
> summary(lm(lnsalesd~as.factor(pair),data=d))
Coefficients:
                   Estimate Std. Error t value Pr(>|t|)
(Intercept)          2.9850     0.1212  24.621 2.72e-12 ***
as.factor(pair)2     1.1250     0.1715   6.562 1.82e-05 ***
as.factor(pair)3     1.1300     0.1715   6.591 1.74e-05 ***
as.factor(pair)4     1.1450     0.1715   6.678 1.52e-05 ***
as.factor(pair)5     1.2800     0.1715   7.466 4.73e-06 ***
as.factor(pair)6     1.4100     0.1715   8.224 1.65e-06 ***
as.factor(pair)7     1.5750     0.1715   9.186 4.77e-07 ***
as.factor(pair)8     1.5850     0.1715   9.245 4.44e-07 ***
as.factor(pair)9     1.6100     0.1715   9.390 3.71e-07 ***
as.factor(pair)10    1.7950     0.1715  10.469 1.05e-07 ***
as.factor(pair)11    1.8100     0.1715  10.557 9.56e-08 ***
as.factor(pair)12    2.0150     0.1715  11.752 2.68e-08 ***
as.factor(pair)13    2.0700     0.1715  12.073 1.94e-08 ***
--
Residual standard error: 0.1715 on 13 degrees of freedom
Multiple R-squared:  0.9474,        Adjusted R-squared:  0.8988
F-statistic:  19.5 on 12 and 13 DF,  p-value: 2.356e-06
```

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How to block?

- Stratification
- Pair-wise matching; two pair-matching

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- If doing randomization inference, block design influences construction of Ω: re-randomization is within blocks.
- Also, type of SE can influence type of bootstrap, e.g. use the block bootstrap (resample at level of blocks) for clustered SEs

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- Finding partners, manage relationships will be common failure point
- Iron law of field experiments: keep it simple. Many ways to fail.
- Getting difficult to publish studies that only examine the efficacy of some intervention: look for substantively interesting treatment and outcome.

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- Beyond methodology, remember to consider:
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